DA19 Project question and a safe plan

Week 7, Show it | Session 19 of 24 | Python for Data Analysis

Today's question. Which three questions can our data answer, and how will we show the answers?

Objective. By the end: you can write a narrow project plan with a question, the data, the measures, the charts and the limits, and have it approved.

> **WHERE THIS IS USED**
> Data teams write a short plan before any code: the question, the data, the charts and what the data cannot say. Which three questions about a topic you love could a table answer?

Before this session: DA18 Does your total match the true total? Clean, merge and reconcile. If you cannot explain the last example, use its error card with the instructor first.

Words for today

Word	What it means	Think of it like this
project question	One question a table can answer with numbers.	Which item sells most, asked of a till roll.
scope	How much the project covers, and what it leaves out.	The fence round a garden.
data dictionary	A list of every column with its meaning and unit.	The key to a map.
limitation	What the data or method cannot tell us.	The part of the map that is blank.
licence	The rules that say how you may use a dataset.	The terms on a borrowed book.
disclosure	A plain statement of any help or AI tool you used.	Writing your sources at the end of an essay.

Block A. From a topic to three questions

A project serves a reader. Start with the topic on your progress map. Decide who needs the answer and what choice it could help with. Review AI_USE_POLICY.pdf now. Any permitted supervised AI help is written in AI_USE_LOG.md, and your own attempt always comes first. No forecasting or model training is required.

Predict. Predict which of these questions the Sunrise Shops data can answer with the taught methods: (a) Which category earns the most at each branch? (b) Why do people like Campus Gate? (c) What will sales be next month? Write your guess on paper first.

Your guess, written before you run it

Run it, then compare. Question (a) can be answered: it needs the sales, products and branches tables, a merge and a group total. Question (b) asks for reasons, and the data has no opinions. Question (c) asks

for a forecast, which this course does not teach. A narrow descriptive question with named columns is the right size.

Investigate.

1. What makes question (a) answerable?
2. How would you narrow question (b) so the data can answer it?

Modify. Write one more question the Sunrise data can answer, and name the columns it needs.

Block B. The plan and the data

Your PROJECT_PLAN.pdf has seven headings: question, reader and use, data source and licence, measures and columns, method, charts, limitations. Start with Sunrise Shops, which is invented and needs no permission. You may use a public real dataset, for example from the Uganda Bureau of Statistics or the World Bank, once the licence has been read at the source, the instructor has approved it, and it holds no personal information.

Predict. Predict which heading is hardest to write, and why. Write your guess on paper first.

Your guess, written before you run it

Run it, then compare. Most learners find limitations hardest, because it means saying what the data cannot show. A good limitation names a missing column or a kind of conclusion the project will not draw: no costs means no profit, invented data means no claim about real shops.

Investigate.

1. Why must the licence be read at the source?
2. What goes in the data dictionary?

Modify. Write the limitations heading for your own plan in two sentences.

Make

Write PROJECT_PLAN.pdf with the seven headings. Start from the topic on your progress map and choose the supplied data file that fits it best. Real public data is used only after you and the instructor have read its licence. Name the data file and its licence or its invented status. List the three questions, the columns each needs, the charts and the limitations. Ask the instructor to approve a narrow scope before you build.

Save your work as DA19. Keep the worked example and put your own attempt below it. Write two sentences: what you changed and how you checked it.

Error card

You see	It means	Fix
The plan promises profit	The data has no costs, so profit cannot be worked out.	Change the measure to revenue and say so in the limitations.
A question needs a column that does not exist	The question asks for something the data never recorded.	Narrow the question to the columns you have, or choose another dataset.

Exit ticket

My plan has three questions that my data can answer, and it says what my data cannot show.

AI use follows AI_USE_POLICY.pdf. In weeks 1 to 6 no AI-generated solutions are used for exercises, labs or build tasks. Practise for 20 to 30 minutes between classes by repeating the Make task with one value changed. If you are stuck for 30 minutes, save the error and bring it to class. A file saved on your laptop has not been shared until you use the route in START_HERE.pdf.